

Equations of the form

$$|f(x)| = |g(x)| \quad (i)$$

or inequalities

$$(ii) |f(x)| \leq |g(x)| ; |f(x)| \geq |g(x)| \quad (iii)$$

should be solved using the fact that

$$|f(x)| = \sqrt{[f(x)]^2}$$

Hence,

$$(i) \Leftrightarrow [f(x)]^2 = [g(x)]^2$$

$$(ii) \Leftrightarrow [f(x)]^2 \leq [g(x)]^2$$

$$(iii) \Leftrightarrow [f(x)]^2 \geq [g(x)]^2$$

Example: solve for x

$$(a) |x-3| = |2x+1|$$

$$(b) |3x-1| \geq |2x|$$

$$(c) \frac{|x-1|}{|x+2|} \not\geq 3$$

$$(a) |x-3| = |2x-1|$$

$$\Leftrightarrow (x-3)^2 = (2x-1)^2$$

$$\Rightarrow x^2 - 6x + 9 = 4x^2 - 4x + 1$$

$$\Rightarrow 3x^2 + 2x - 8 = 0$$

$$\Rightarrow x = \frac{-2 \pm \sqrt{4 + 4 \times 8 \times 3}}{6} = \frac{-2 \pm 10}{6}$$

$$\Rightarrow x = -\frac{12}{6} = -2 \quad \text{or} \quad x = \frac{8}{6} = \frac{4}{3}$$

$$\Rightarrow x \in \left\{-2, \frac{4}{3}\right\}$$

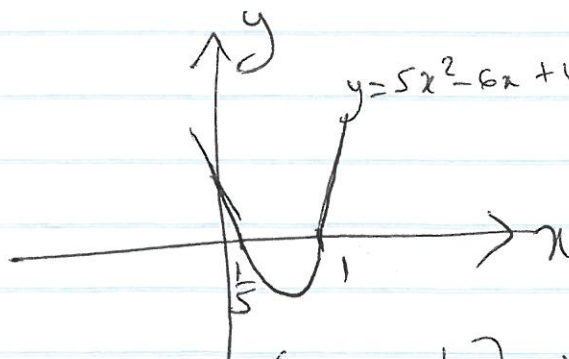
$$(b) |3x-1| \geq |2x|$$

$$\Leftrightarrow (3x-1)^2 \geq (2x)^2$$

$$\Rightarrow 9x^2 - 6x + 1 \geq 4x^2$$

$$\Rightarrow 5x^2 - 6x + 1 \geq 0$$

$$\Rightarrow 5x^2 - 6x + 1 = 0 \Rightarrow x = \frac{6 \pm \sqrt{36 - 20}}{10}$$



$$x = \frac{6 \pm 4}{10}$$

$$x \in \left\{\frac{1}{5}, 1\right\}$$

$$\Rightarrow x \in (-\infty, \frac{1}{5}] \cup [1, +\infty)$$

$$(c) \frac{|x-1|}{|x+2|} \leq 3 \Leftrightarrow |x-1| \leq 3|x+2|$$

since $|x+2| > 0$
and $x \neq -2$

$$\Rightarrow (x-1)^2 \leq 3(x+2)^2$$

$$\Rightarrow x^2 - 2x + 1 \leq 3(x^2 + 4x + 4)$$

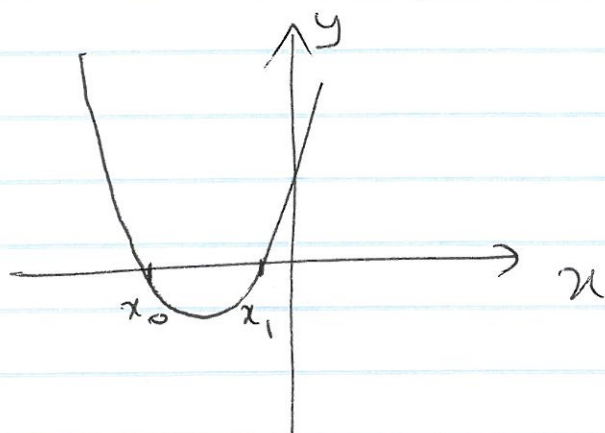
$$\Rightarrow 0 \leq 2x^2 + 14x + 11$$

Consider $2x^2 + 14x + 11 = 0$, $\Delta = (14)^2 - 8 \times 11 = 108$

$$x = \frac{-14 \pm \sqrt{108}}{4}$$

$$x_0 = \frac{-14 - \sqrt{108}}{4}$$

$$x_1 = \frac{-14 + \sqrt{108}}{4}$$



$$\Rightarrow x \in (-\infty, x_0) \cup (x_1, +\infty)$$